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SIMULATING FILTRATION PROCESSES USING A NETWORK MODEL

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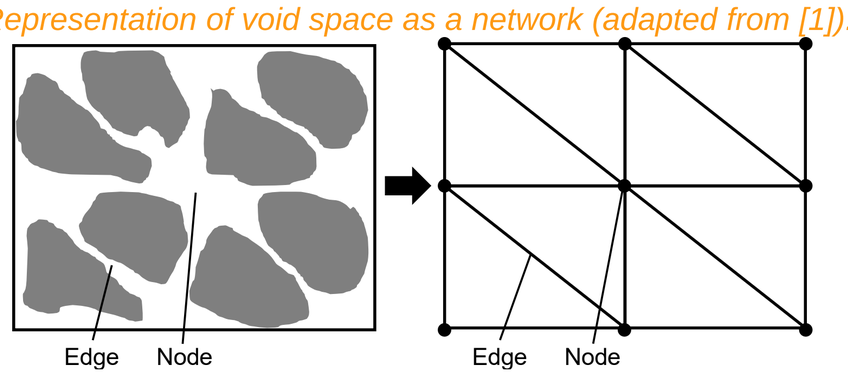
INTRODUCTION

In hydraulic earth structures, granular filters are employed adjacent to base material to prevent erosion. Granular filter design rules have generally been empirical; therefore, a need arises to understand filtration processes at a particulate level. Although particle-based methods (e.g. coupled DEM-CFD) offer perhaps the most complete account of the filtration processes, the increased computational cost limits the number of particles that can be considered simultaneously. Network models provide an attractive alternative that allows computations to be performed at pore-scale.

NETWORK MODEL

A network model in its simplest form is a set of nodes interconnected with edges. The void space of granular filters consists of pores that are linked by narrow constrictions, which can be idealised as nodes and edges respectively (see Figure 2).

Figure 1: Representation of void space as a network (adapted from [1]).



The research proposes a network model that incorporates the following:

1. Accurate void space structure (Figure 2).
2. Simultaneous simulation of multiple base particles, introduced based on a specified PSD, within the void space.
3. Realistic fluid flows, which allow the application of 'flow-biased probabilities' proposed by [1] (Figure 3).
4. Simulation of filtration by edge and node clogging (Figure 3).

Figure 2: The research considers (a) the virtual filter samples created using the DEM by [2], and (b) the application of the DTM as documented in [3].*

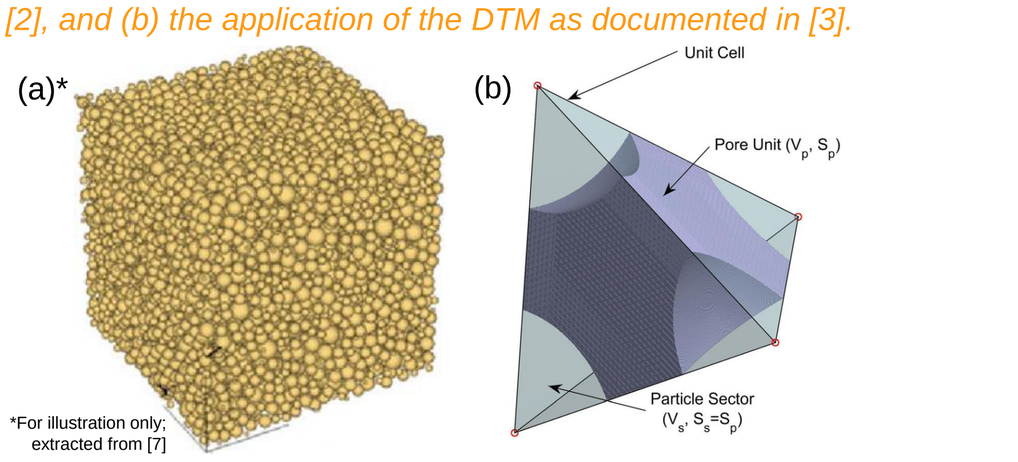
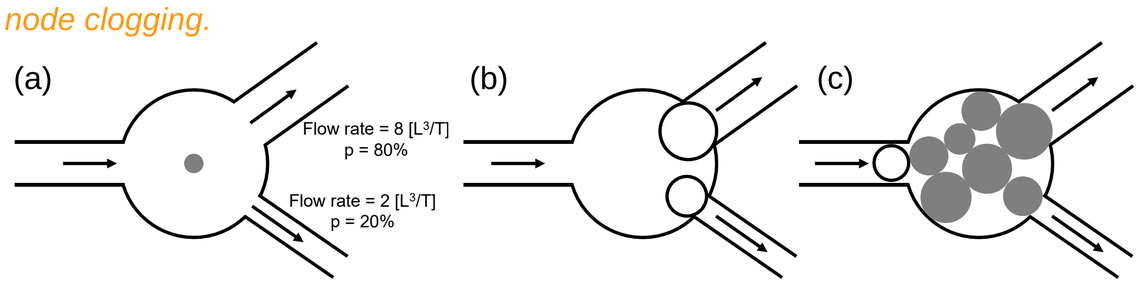


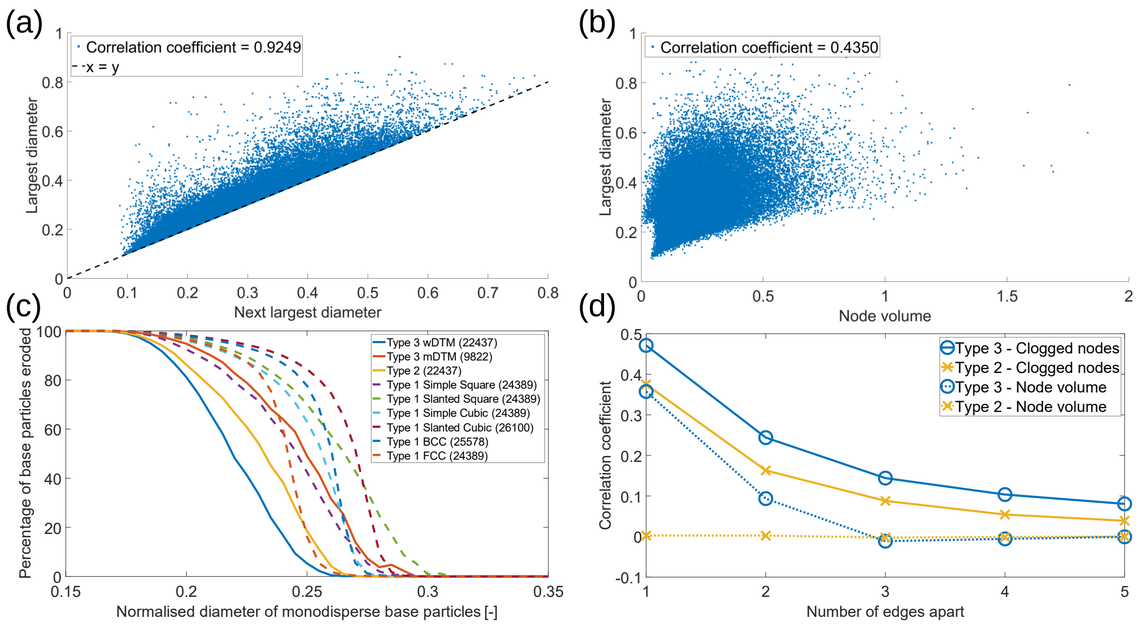
Figure 3: Simulation of (a) flow-biased probabilities, (b) edge clogging, and (c) node clogging.



ACCURATE VOID SPACE STRUCTURE

Many of the previous network models made idealising assumptions that often oversimplify the structure of the void space, i.e. (1) regular lattices with (2) geometry properties randomly distributed. The comparison of results based on networks that match the actual void space to varying degree demonstrates the importance of modelling accurate void space structure.

Figure 4: (a & b) Spatial correlation exists in random sphere packings, (c) the filter effectiveness is directly influenced by the network characteristics, and (d) two-point spatial correlation of node properties.



INSIGHTS INTO FILTRATION PROCESSES

Simulations using the adopted network model allow the fundamental mechanics to be observed and analysed. Two examples are shown below.

Figure 5: Larger base particles follow a smaller number of common erodible paths. (a) Base particle size d = 0.250, (b) d = 0.265, and (c) d = 0.280.

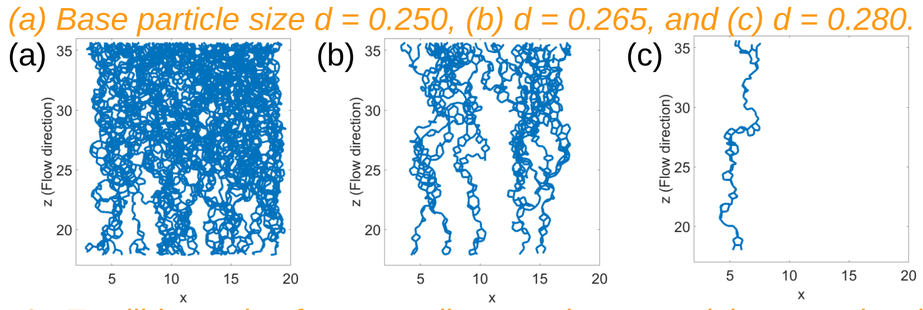
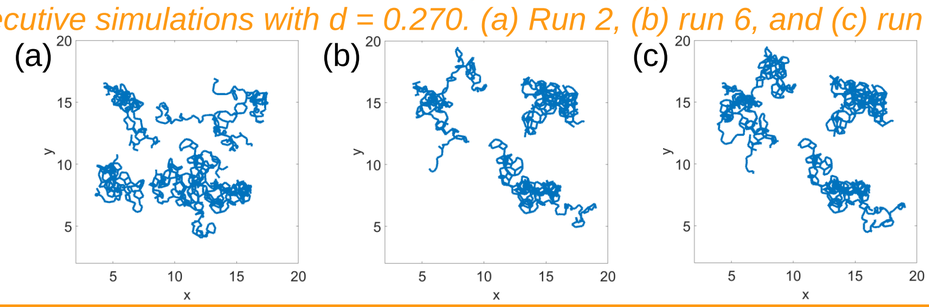


Figure 6: Erodible paths for monodisperse base particles remain similar for consecutive simulations with d = 0.270. (a) Run 2, (b) run 6, and (c) run 10.

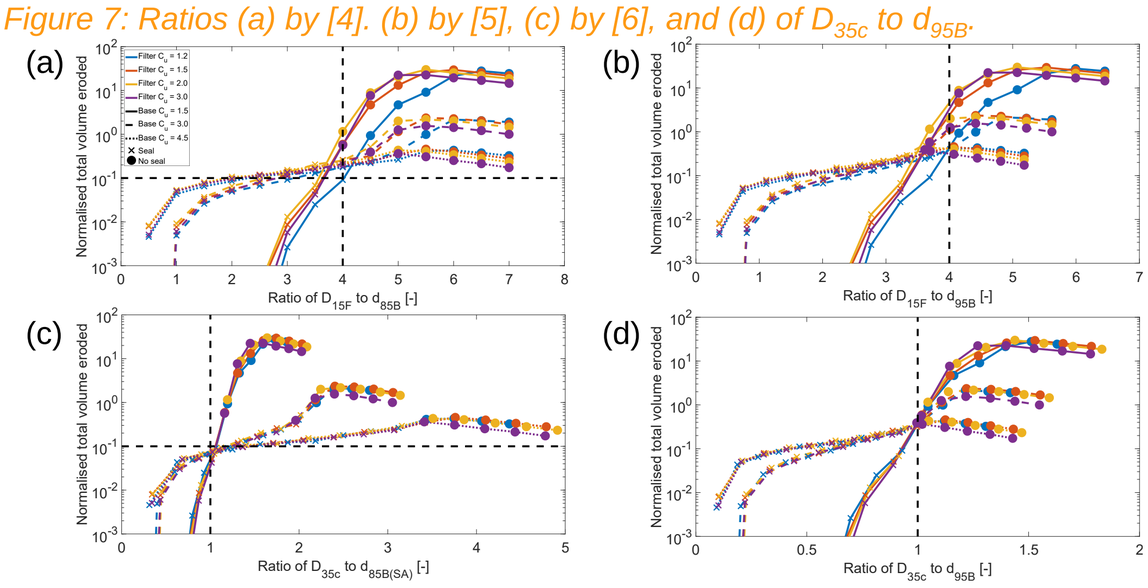


GRANULAR FILTER DESIGN RULES

Simulations based on 504 filter-base combinations are used to assess design frameworks proposed by [4], [5] and [6]:

- The use of $d_{85B(SA)}$ [6] instead of d_{85B} [4] ensures the presence of sufficient number of larger base particles, thus best divides (1) seal with no erosion and (2) seal with some erosion.
- The use of d_{95B} [5] forms better boundary between (2) seal with some erosion and (3) no seal with continuous erosion, since it is the largest base particles that govern the eventual sealing of filters.
- The use of D_{15F} [4] [5] results in limited unifying effect for varying filter characteristics; the use of D_{35c} [6] provides higher explanatory power.
- The adoption of 35th percentile (100 minus percolation threshold) of the filter CSD as the controlling constriction size is supported by the percolation threshold remaining similar for varying filter characteristics.

Figure 7: Ratios (a) by [4]. (b) by [5], (c) by [6], and (d) of D_{35c} to d_{95B}.



CONCLUSION

The research has proposed a network model that incorporates two novel features, i.e. accurate void space structure and realistic fluid flows. The research is able to demonstrates the significance of modelling realistic void space structure, to provide insights into the fundamental mechanics of filtration processes, and to assess the accuracy of established design criteria.

The proposed model consistently allowed larger base particles to be eroded than that observed in experiments. This can be examined in future research to improve accuracy. Two possible applications of the network model include:

- Investigate how filter anisotropy affects filtration behaviours.
- Examine how filters can be optimised by varying its particle size with depth.

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